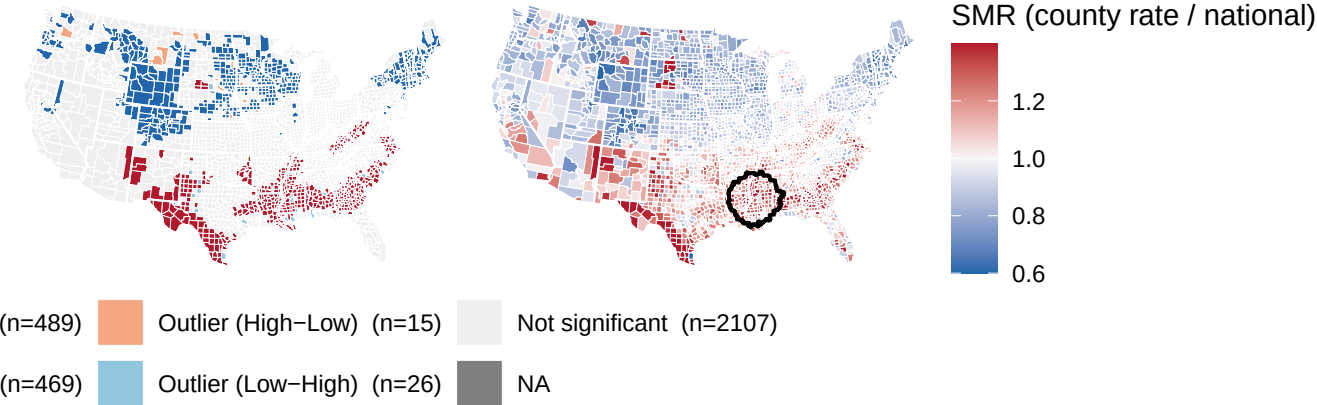


Two answers to 'where is the cluster?' – same county diabetes data, real polygons

Global Moran's I (LISA) – autocorrelation view **Kulldorf spatial scan – detection view**

county classified · Global Moran's I = 0.661 (p=0.016) · Cluster: 145 counties, RR=1.34, p=0.010 (outlined)

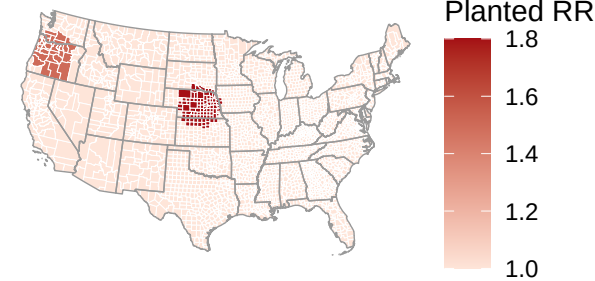


Cluster-detection performance against known ground truth

Two planted clusters, Poisson-simulated counts

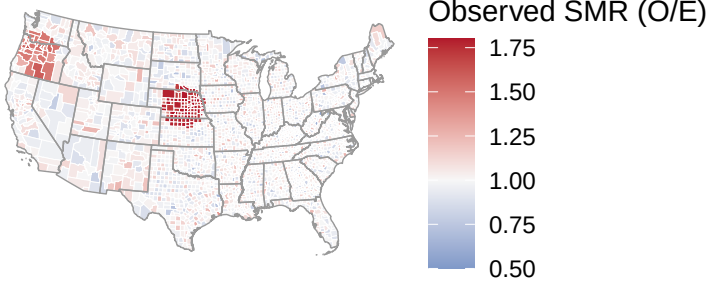
TRUTH: two planted clusters

NE (RR=1.8, 100 cos), OR (RR=1.5, 34 cos)



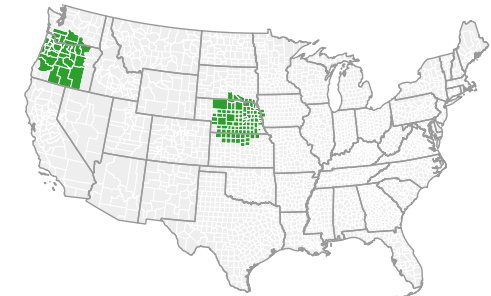
Simulated observed data (SMR)

What each detector actually sees



Kulldorff scan – detection outcome LISA hot-spots – detection outcome

Sens=1.00 Spec=1.00 PPV=1.00 RR=1.71



(n=134)

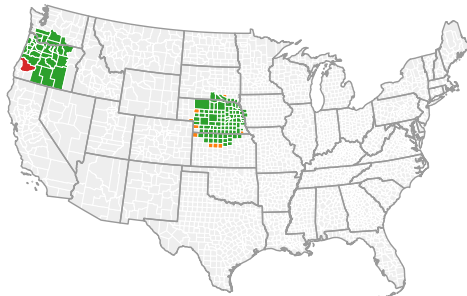
FN (n=0)

FP (n=0)



TP (n=134)
TN (n=2964)

Sens=0.99 Spec=1.00 PPV=0.93 RR=1.65



FN (n=1)

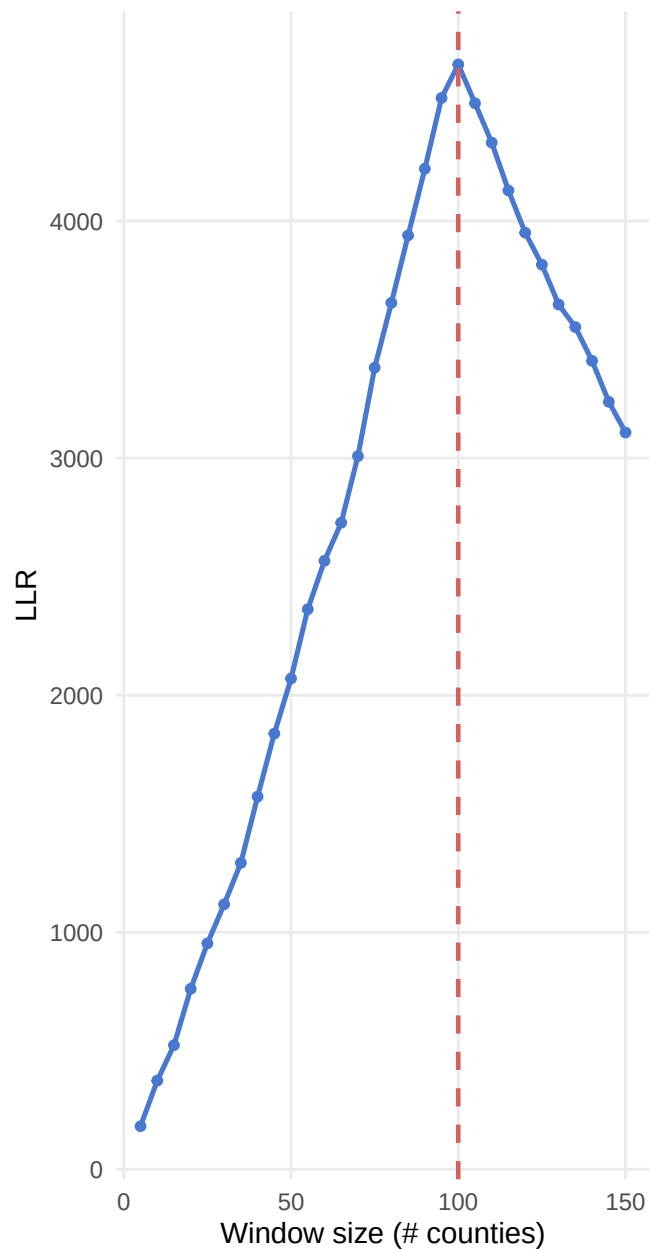
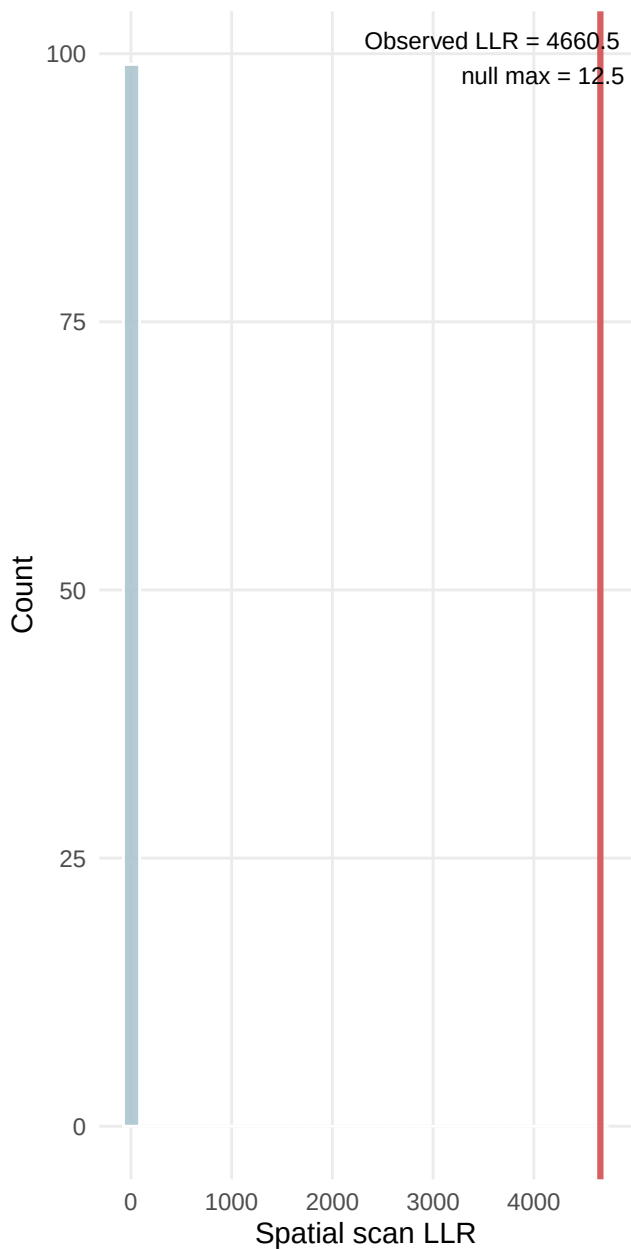


FP (n=10)



TN (n=2964)

Monte Carlo null distribution (B=99) $p=0.01100$ by scanning window size (optimal)

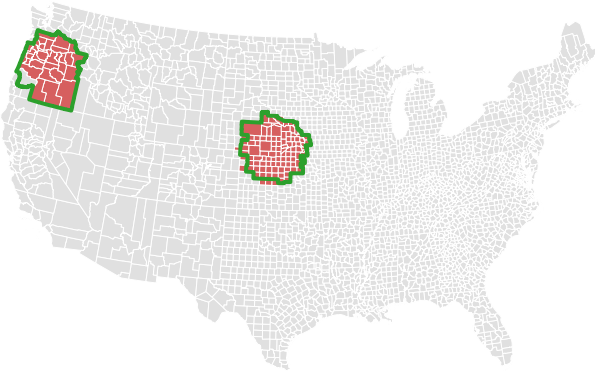


Cluster detection: LISA vs spatial scan statistic

Planted ground truth outlined in green

LISA (HH, $p<0.05$)

143 flagged | 133 true positives | 10 false positives



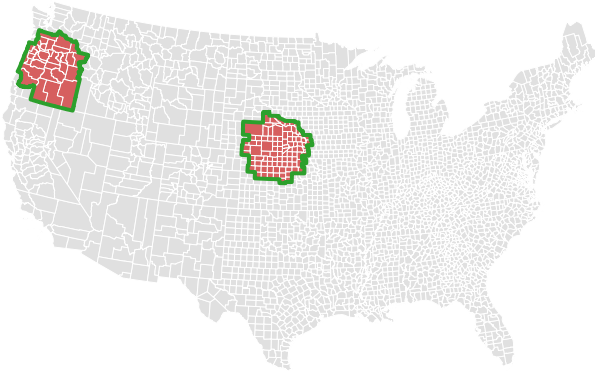
Flagged



Not flagged

Spatial scan statistic

134 flagged | 134 true positives | 0 false positives



Flagged



Not flagged